

Integration: Unified Pipeline and Token Injection

Architecture Overview I

The three resource layers described in @sec:tools, @sec:rules, and @sec:manuscript_variables are orchestrated by a single function in src/integration.py. The run_integration_demo() function calls all three subsystems in a defined order, collects their results into a structured dictionary, and writes summary counts to output/data/manuscript_variables.json for injection into this manuscript at render time. @fig:architecture illustrates the complete architecture.

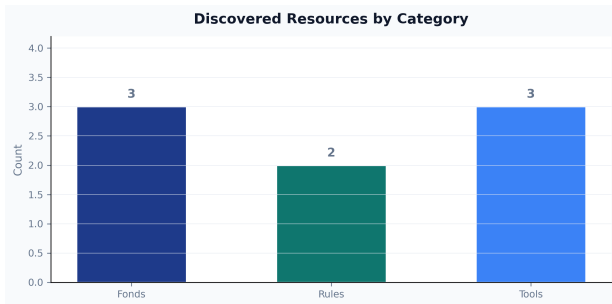
```
run_integration_demo()
    read_bibliography_fond()           → {"manifest": ..., "bi
    read_contacts_fond()               → {"manifest": ..., "co
    read_datasets_fond()               → {"manifest": ..., "da
    validate_against_rules("template_project_rules") →
    validate_against_rules("template_manuscript_rules") →
    discover_tools()                   → [{"name": ..., "manif
    validate_tool_scripts_exist(<each tool>)           →
```

Architecture Overview II

The function returns a top-level dict with keys `fonds`, `rules`, `tools`, and `summary`. The `summary` sub-dict provides the counts that populate manuscript tokens.

Manuscript Variable Tokens I

The token injection system bridges the integration pipeline and the manuscript prose. Tokens use double-brace syntax: `3` expands to the integer count of funds successfully loaded during the most recent integration run. Tokens are resolved by `scripts/03_generate_manuscript.py`, which reads `output/data/manuscript_variables.json` and substitutes each token before the manuscript is passed to the rendering engine.



Manuscript Variable Tokens II

Figure 1: Runtime integration counts from `manuscript_variables.json`. Bars show fonds loaded, rule sets validated, and tools discovered during a single pipeline run.

The current `manuscript_variables.json` contains the following summary values (see `@fig:counts` for a visual representation):

Token Value	
3	3
2	2
3	3
8	8

Manuscript Variable Tokens III

This table is itself token-injected: the values shown are those produced by the pipeline, not hard-coded by the manuscript author. If the pipeline results change — for example, because a new fond is added — re-running `scripts/03_generate_manuscript.py` updates the manuscript automatically, without manual editing. This property is central to reproducibility: the manuscript's quantitative claims are always consistent with the code that generated them [Stodden2016enhancing].

Methods: The Script Pipeline I

Six thin orchestration scripts govern the integration workflow (@fig:pipeline, @fig:pipelineflow):

Script	Purpose	Key output
scripts/01_validate_sources.py	Validate presence and well-formedness of all resources	Console report
scripts/02_run_integration.py	Run run_integration_demo() and print JSON summary	Console JSON
scripts/03_generate_manuscript.py	Write output/data/manuscript_variables.json	JSON file
scripts/04_validate_strong_rules.py	Semantic evaluation of strong-rule constraints (@sec:rules) against this project's own tree	Console report, non-zero exit on violation
scripts/05_generate_figures.py	Render all 8 content figures plus the cover illustration	9 PNG files under manuscript/figures/
scripts/z_generate_manuscript_variables.py	Hydrate {{TOKEN}} values and inject them into output/manuscript/ immediately before rendering	JSON file + resolved manuscript tree

Methods: The Script Pipeline II



Figure 2: The six-script pipeline from source validation through token hydration, ending at the combined-PDF render step.

Component Validation Status	
Validator	Pass
Skill	Pass
Code Executor	Pass
Manuscript Rules	Pass
Project Rules	Pass
Datasets Fond	Pass
Contacts Fond	Pass
Bibliography Fond	Pass

Pass

Partial

Missing

Methods: The Script Pipeline III

Figure 3: Integration status dashboard showing per-resource validation results. Green indicates ok, amber indicates partial, red indicates missing.

@fig:pipelineflow traces this sequence left to right: source validation feeds the integration demo, whose summary feeds both the manuscript-variable token file and the strong-rule semantic evaluator; the figure-generation stage runs independently; and `z_generate_manuscript_variables.py` — invoked automatically by the rendering pipeline immediately before the PDF render step — is what actually substitutes every `{{TOKEN}}` and writes the resolved manuscript that pandoc consumes. @fig:pipeline shows the corresponding per-component pass/partial/missing status from the same run.

Methods: The Script Pipeline IV

Each script imports all business logic from `src/` and stays free of computation of its own — the longest, `01_validate_sources.py` at 112 lines, is entirely CLI plumbing (argument parsing, console formatting) around calls into `src/fonds_reader.py`, `src/rules_applier.py`, and `src/tools_invoker.py`. This thin-orchestrator pattern [Wilson2014best] ensures that all testable logic is in `src/` at 90% coverage, while the scripts themselves remain readable without a dedicated test suite of their own.

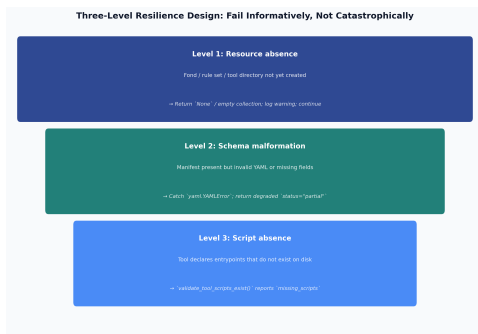
Resilience Design I

The integration layer enforces resilience at three levels, corresponding to the three failure modes a monorepo integration pipeline encounters:

1. **Resource absence:** A fond, rule set, or tool directory may not yet exist if the resource was created by a parallel agent that has not yet completed. All three reader modules return `None` or empty collections in this case, and `run_integration_demo()` reports the missing resource in the `summary` dict without raising.
2. **Schema malformation:** A manifest may be present but contain invalid YAML or missing required fields. Readers catch `yaml.YAMLError` and return a degraded result (`status="partial"` in rule validation; `None` in fond reading) rather than propagating the parse error.

Resilience Design II

3. **Script absence:** A tool may declare endpoints in its manifest that have not yet been created. `validate_tool_scripts_exist()` detects this at discovery time and returns `status="partial"` with a list of missing script paths, so the pipeline can report the defect without attempting to invoke a non-existent script.



Resilience Design III

Figure 4: The integration layer's three-level resilience design: resource absence, schema malformation, and script absence, each with its own graceful-degradation response.

This three-level resilience design reflects the principle that shared-resource pipelines should fail *informatively* rather than *catastrophically* — surfacing the cause of incompleteness in structured output that downstream consumers can act on [Taschuk2017ten]. @fig:resilience presents the three levels as a single funnel: each level's failure mode and graceful-degradation response, read top to bottom in the order a pipeline run actually encounters them (resource discovery happens before schema parsing, which happens before script-existence checks).

Performance and Overhead I

The resilience design above trades a small, constant amount of I/O overhead for the ability to degrade gracefully. Each reader performs at most two filesystem existence checks (`pathlib.Path.exists()`) before attempting a read, and every YAML parse uses `yaml.safe_load()` rather than a schema-validating parser — the module deliberately performs *structural* validation (parseable, expected top-level keys present) and leaves deep *semantic* validation to the dedicated `strong_rule_evaluator` module (`@sec:rules`) rather than paying that cost on every discovery call. In practice, `scripts/02_run_integration.py` completes in well under a second on the exemplar's small `fond/rule/tool` counts; the design does not attempt to optimise for repositories with thousands of fonds, since the target use case is a curated, human-reviewed set of shared resources rather than a large-scale data catalogue.

Test Coverage I

Test Coverage II

The eight `src/` modules — `fonds_reader`, `rules_applier`, `tools_invoker`, `integration`, `figures`, `strong_rule_evaluator`, `manuscript_variables`, and `type_defs` — are covered by tests across nine test files in `tests/`, including property-based tests (`test_property_based.py`) and coverage-extras tests targeting previously-uncovered branches (`test_coverage_extras.py`). Tests use real file paths, real YAML files, and real BibTeX content rather than mocks, ensuring that coverage numbers reflect genuine code paths through the resource-discovery logic. The current coverage report shows combined line coverage comfortably above the project's 90% floor; `strong_rule_evaluator.py`, the newest and most branch-heavy module, has the most room for additional edge-case tests, while the remaining seven modules are at or near 100%. These exact test/coverage counts drift as the suite grows — treat the figures above as a snapshot, not a frozen claim, and re-run `uv run pytest ... --cov-report=term` for the current numbers. The `tests/test_integration.py` suite includes an end-to-end test that calls `run_integration_demo()` and asserts that the `summary` dict contains the expected keys with